In Proceedings IROS, 2-6 August 2005, Edmonton

Experimental and Simulation Results of Wheel-Soil Interaction for Planetary Rovers

Robert Bauer
Department of Mechanical Engineering
Dalhousie University
Halifax, Nova Scotia, B3H 1B6, Canada

robert.bauer@dal.ca

Winnie Leung, Tim Barfoot
Controls and Analysis
MDA Space Missions
Brampton, Ontario, L6S 4J3, Canada

{wleung, tbarfoot }@mdrobotics.ca

***Abstract* – The ability to predict rover locomotion performance is critical during the design, validation and operational phases of a planetary robotic mission. Predicting locomotion performance depends on the ability to accurately characterize the wheel-soil interactions. In this research, wheel-soil interaction experiments were carried out on a single-wheel testbed and the results were compared with a single-wheel dynamic computer simulator which was developed in Matlab and Simulink's SimMechanics toolbox using a commercially-available wheel-soil interaction computer model called AESCO Soft Soil Tire Model (AS^{2}TM). Two different tire treads were used and compared in this study. There is good agreement between experimental and simulation results for wheel sinkage as a function of slip ratio; however, more investigation is needed to understand the differences observed for the drawbar pull and motor torque results.**

***Index Terms* – wheel-soil interaction, dynamic simulation, planetary rovers**

I. Introduction

Rovers will continue to be an essential element to future planetary exploration missions. These missions will require rovers to travel over challenging terrain to achieve ambitious scientific objectives. The ability to predict rover locomotion performance is critical during the design, validation and operational phases of a planetary robotic mission. Accurate prediction and optimization of rover locomotion performance requires an understanding of the wheel-soil interactions.

Several approaches have been proposed to model sandy soil. One computationally expensive approach is to model every soil grain and its interaction with neighboring grains. Grand *et al.* [1] and Andrade [2] use a planar macroscopic approach to model the global behavior of sandy soil based on finite elements where the soil is divided in column cells which interact with each other. Several analytical models have also been developed [3][4].

This research uses Matlab and Simulink's SimMechanics toolbox[5] in conjunction with a commercially-available wheel-soil interaction model called AESCO Soft Soil Tire Model (AS^{2}TM)[6][7][8]. This model provides extended versions of the traditional analytical methods introduced by Bekker [9] and Wong [10]. The vertical and horizontal deformations in the wheel-soil interaction are separated and described by pressure-sinkage characteristics and shear-tension-displacement characteristics, respectively.

The pressure-sinkage relationship, as proposed by Bekker [9], is described as:

$$p = \left(\frac{k_c}{b} + k_\varphi \right) z^n \tag{1}$$

where p is the pressure, b is the smaller dimension of the contact path/width of the rectangular contact area, z is the sinkage, and n, k_c and k_φ are the pressure-sinkage parameters. The parameter n is called the exponent of sinkage while k_c and k_φ are called the cohesive and frictional moduli of deformation, respectively.

The maximum shear stress can be described by either the Coulomb rule:

$$\tau_{\max} = c + p \tan(\varphi) \tag{2}$$

or by adhesion between the wheel and the soil as follows:

$$\tau_{\max} = p\mu \tag{3}$$

where c is the cohesion of the soil, φ represents the internal friction angle of the soil, and μ is the friction coefficient. AS^{2}TM chooses the minimum between the adhesion and the internal soil friction to calculate the maximum shear stress [6].

The shear-stress to shear-deformation relationship, as proposed by Janosi and Hanamoto [11], is described by

$$\tau = \tau_{\max} \left(1 - e^{-\frac{j}{K}} \right) \tag{4}$$

where τ is the shear stress, j is the shear deformation, and K is the tangent modulus of horizontal shear deformation (or slip coefficient).

In order to model the rigid wheels used in this research, AESCO added a rigid wheel option to AS^{2}TM. With this option the local pressure and local shear displacement under a rigid wheel is used to compute the local stress. Integrating the local shear stress over the contact area provides the lateral and longitudinal forces, while integrating the local pressure (i.e. normal stress) along the contact area provides the vertical reaction force.

Several important effects are modeled in AS^{2}TM. For example, in the case of a rigid wheel, the rolling resistance is a result of plastic soil deformation as well as slip sinkage.

AS^{2}TM accounts for the tire tread by considering the grouser height and ratio between positive and negative portions of the tread. AS^{2}TM also has a multipass option wherein the front wheels create a track in which the rear wheels follow. This option considers both deformation and precompaction of the soil behind each wheel.

II. Wheels tested

A cylindrical wheel was tested in this research. This wheel was designed so that the number of grousers or lugs on the wheel could be easily changed from 9 to 18 by attaching different plates to the surface of the wheel. Fig. 1 shows the two grouser configurations used in this research.

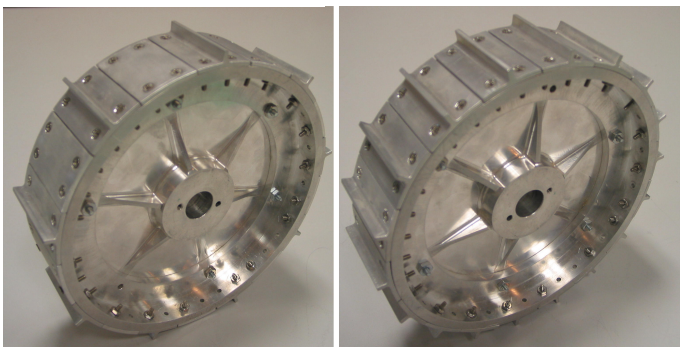

Fig. 1: Cylindrical Wheel with 9 and 18 grouser plates attached

III. Experimental apparatus

To validate the wheel-soil interaction model using these two wheel treads, experiments were performed on the Massachusetts Institute of Technology (MIT) Field and Space Robotics Laboratory's Wheel-Terrain Characterization Testbed as shown in Fig. 2. This testbed consists of a wheel carriage that can translate both horizontally and vertically. Potentiometers sense carriage motion. A torque sensor and motor are attached to the wheel and a force/torque transducer is located on the wheel carriage above the wheel. Controlling the translational velocity of the wheel carriage and the angular velocity of the wheel enables one to control the slip ratio which is defined as:

$$i = 1 - \frac{v}{\omega r} \tag{5}$$

where v is the component of wheel carrier velocity in the horizontal direction, ω is the wheel rotational speed, and r is the wheel radius.

With this setup, the wheel-soil interaction forces and torques (and the corresponding slip ratio) can be measured. The directions for positive sensor force and torques are superimposed in Fig. 2 and forward motion corresponds to the wheel moving to the left of this figure.

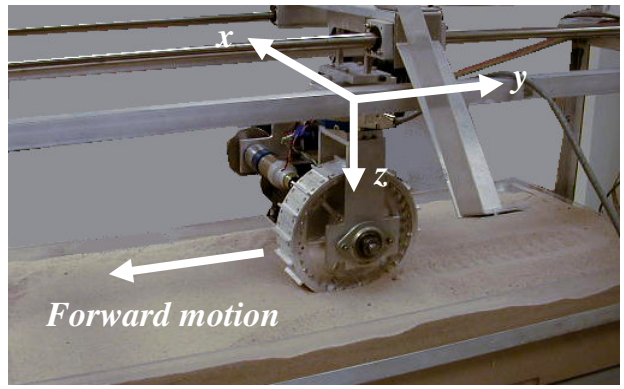

Fig. 2: Wheel-Terrain Characterization Testbed at MIT's Field and Space Robotics Laboratory

A dry sandy soil was used in the testbed and a series of experiments were conducted to characterize the soil parameters. Soil density ρ was measured using an electronic balance. Flat-plate sinkage experiments, as shown in Fig. 3a), were performed similar to the method described by Bekker [9] and Wong [10] using flat plates constructed to correspond to the size of the test wheel. The results of this analysis provided estimates of the pressure-sinkage parameters n, k_c and k_φ. It should be noted that flat-plate sinkage experiments are normally carried out with an apparatus which allows larger vertical loads than those achievable with the Wheel-Terrain Characterization Testbed. Larger loads would induce higher sinkage measurements and improve the accuracy of the calculated n, k_c and k_φ soil parameters. Alternatively, improving the resolution of the potentiometer used on the testbed to measure sinkage would also likely improve the confidence in the calculated soil parameters.

The shear-deformation modulus K was determined using the apparatus shown in Fig. 3b). The apparatus is similar to those used for the standard direct shear test in Civil engineering where known horizontal displacements are imposed at the interface while the vertical displacement is measured [12]. The internal friction angle φ was determined from the slope of the soil when piled as shown in Fig. 3c).

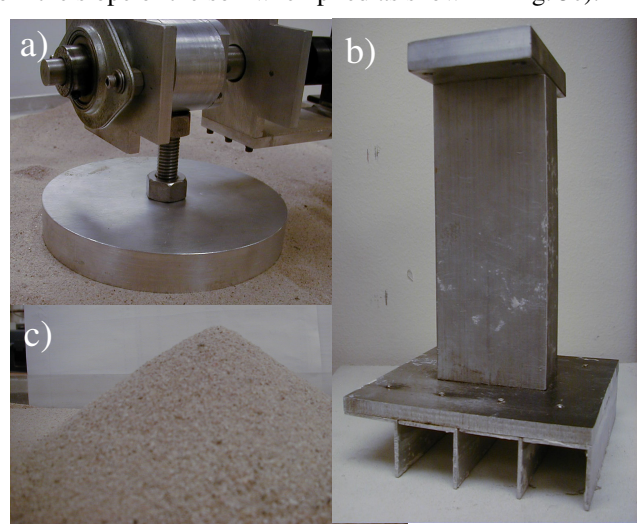

Fig. 3: Characterizing Soil Parameters

Table 1 summarizes the soil parameters measured from these experiments.

Table 1: Measured Soil Parameters

Parameter	*Value*
ρ	1.605 g/cm^3
n	1.1
k_c	0.65 N/cm$^{(n+1)}$
k_φ	3.35 N/cm$^{(n+2)}$
K	1.5 cm
φ	32°

IV. Experimental Results

Fig. 4 and Fig. 5 plot a sample of the experimental results showing the measured wheel torque as a function of time with a slip ratio of 0.24 for the two different wheel treads. A mean value has been superimposed on the plots as well as 95% confidence intervals for the last two seconds of data when steady state has been achieved.

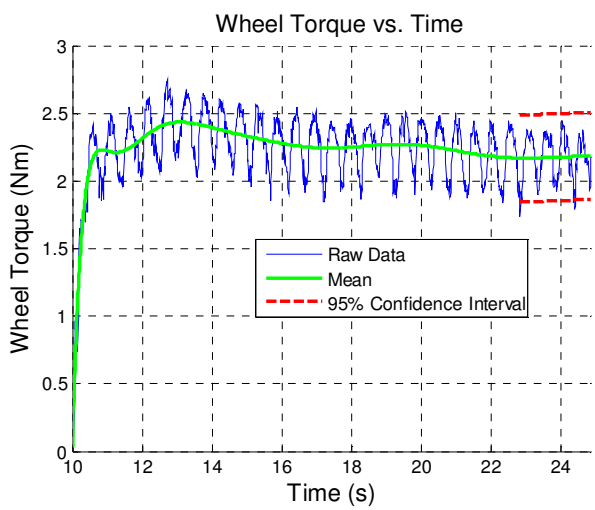

Fig. 4: Measured Wheel Torque vs. Time, 18 grousers

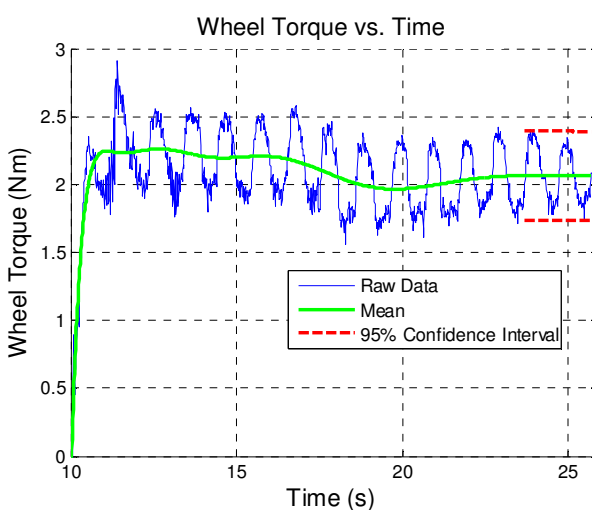

Fig. 5: Measured Wheel Torque vs. Time, 9 grousers

The observed oscillations in the sinkage, force, and torque data are due to the individual grousers interacting with the soil. Doubling the number of grousers on the wheel has the effect of doubling the number of oscillations in the measurements.

For each tire tread, experiments were carried out for the following 7 slip ratios: 0.04, 0.15, 0.24, 0.35, 0.46, 0.57, 0.66. For each slip ratio, at least 3 trials were performed to provide an indication of the repeatability of the experiments. The trials for each experiment were merged into a single dataset and the last two seconds of each merged dataset was averaged to obtain the steady-state mean values. These data were calculated for each of the two tire treads and the experimental results are plotted in Fig. 6, Fig. 7 and Fig. 8 as a function of slip ratio.

Fig. 6 shows that the measured sinkage vs. slip ratio data are very similar for both wheel treads at slip ratios below 0.5. It is evident from Fig. 7 that doubling the number of grousers increases the drawbar pull F_y by approximately 30%. The negative values of the measured sensor forces F_y in Fig. 7 are consistent with the sensor coordinate frames shown in Fig. 2. The motor torques in Fig. 8 are generally slightly larger for the wheel with 18 grousers as expected given the higher drawbar pull forces F_y.

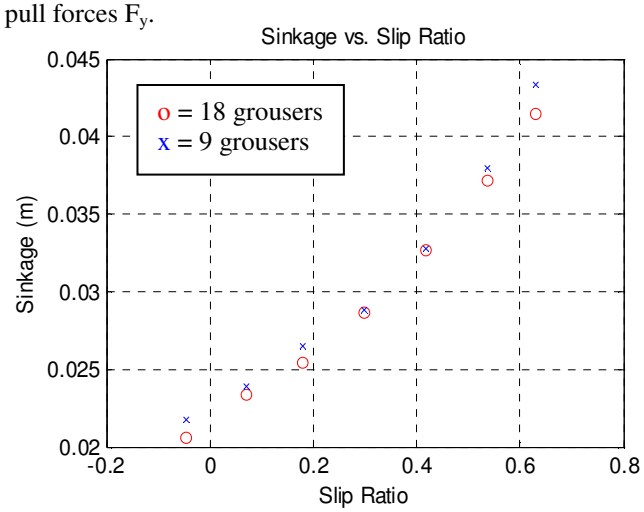

Fig. 6: Measured Sinkage vs. Slip Ratio, 9 and 18 grousers

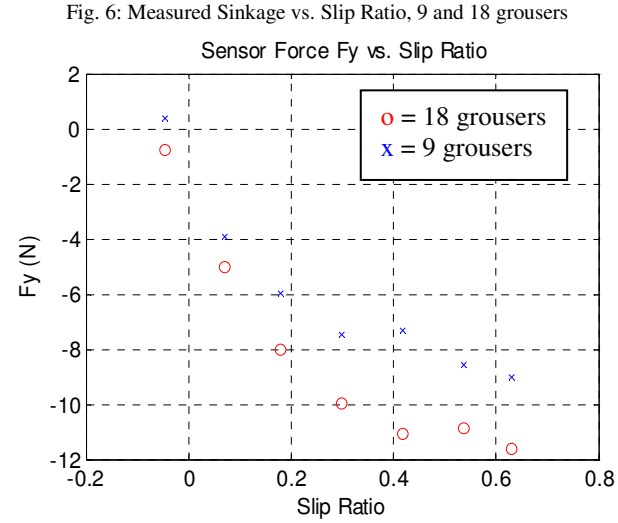

Fig. 7: Measured Drawbar Pull vs. Slip Ratio, 9 and 18 grousers

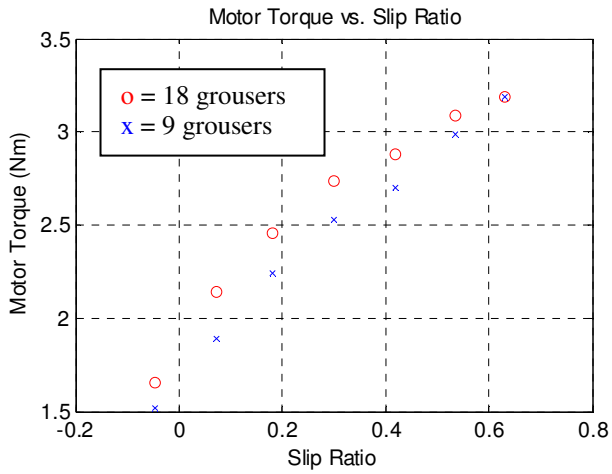

Fig. 8: Measured Motor Torque vs. Slip Ratio, 9 and 18 grousers

V. Comparing Experimental Results with Simulation Results

In order to compare the above experimental results with the AS^{2}TM soft-soil tire model, the soil parameters associated with this model need to be tuned.

A. Tuning of Soil Parameters used in the Model

There are several soil parameters used in AS^{2}TM that need to be determined experimentally. These include the density ρ of the soil, n, k_c and k_φ as defined in equation (1), the cohesion c, friction angle φ, slip coefficient K, stiffness c_B, damping b, *slippery* (maximum friction coefficient for the surface), *grip* (maximum friction coefficient for the tire), *compaction capability* (used for multipass), *rolling resistance correction* (to account for effects such as bulldozing), and *shear offset* (used for sandy soils). While the first 7 of these parameters are traditional soil parameters, the remaining parameters are less conventional and can be tuned using experimental data.

It is challenging to tune these parameters because each tuning parameter often influences more than one measured system response. Based on experience gained by working with the experimental and simulation data, the following tuning approach was developed as part of this research:

1. start with *shear offset=0* and tune k_φ to obtain the correct sinkage at small slip ratios
2. adjust c_B to modify the drawbar pull F_y
3. adjust the *shear offset* so that sinkage predictions agree with experimental results for high slip ratios

During the tuning of these parameters, the following general trends were observed:

- as *rolling resistance correction* increases, *Fy* becomes positive at high slip ratios
- as k_φ increases, the sinkage decreases
- as c_B decreases, sinkage decreases, |*Fy*| increases, and |*Motor Torque*| increases
- as k_c increases, |*Fy*| increases and sinkage decreases
- as c increases, |*Fy*| increases and |*Motor Torque*| increases
- increasing the *compaction capability* parameter reduces the subsequent sinkage predictions as wheels repeatedly pass over the track. Note that *compaction capability* only affects multipass cases.

Table 2 summarizes the soil parameters tuned using this procedure on experimental data from the wheel with 18 grousers. The damping b was assumed to be very high and the cohesion c was assumed to be negligible for the dry sandy soil.

Table 2: Tuned Soil Parameters

Parameter	*Value*
ρ	1.605 g/cm^3
n	1.1
k_c	0.65 N/cm$^{(n+1)}$
k_φ	1.80 N/cm$^{(n+2)}$
K	1.5 cm
c	0
φ	32°
b	4000 Ns/m
c_B	1500 N/cm^3
slippery	0.35
grip	1.1
shear offset	0.16 cm
rolling resistance correction	0.05
compaction capability	2.3

Note that the *compaction capability* parameter was tuned as a last step using multipass experimental data where, after the first pass, the wheel was allowed to pass through its track an additional two times. Also note that AS^{2}TM uses the smallest value between the *slippery* and *grip* parameters for the friction-related calculations.

B. Comparison of Simulation and Experimental Results

Fig. 9 and Fig. 10 compare the resulting experimental and simulation data as a function of slip ratio for wheel treads with both 18 and 9 grousers. Note that for each experimental data point, the 95% confidence interval is plotted. In both cases, the sinkage relationship is accurately modeled. At slip ratios beyond 0.5 the predicted drawbar pull F_y begins to decrease while the measured response levels off; however, the simulation results generally lie within the calculated 95% confidence intervals. A leveling off or decrease in the drawbar pull can be explained physically because the increase in horizontal force due to a higher slip ratio can be offset by an increase in the rolling resistance due to the higher sinkage of the wheel.

Fig. 11 plots the results of the multipass experimental and simulation data as a function of the number of passes for the wheel with 9 grousers. A slip ratio of 0.24 was selected for all of the multipass experiments. In the first pass, the wheel created a fresh track while in the second and third passes, the wheel went over the existing track. From Fig. 11 it is evident that AS^{2}TM is able to capture the sinkage measurements accurately.

Fig. 11 also shows the measured and predicted drawbar pull F_y for the multipass case. The first pass results are consistent with those shown in Fig. 10 for a slip ratio of 0.24. The second and third pass results are within the 95% confidence intervals.

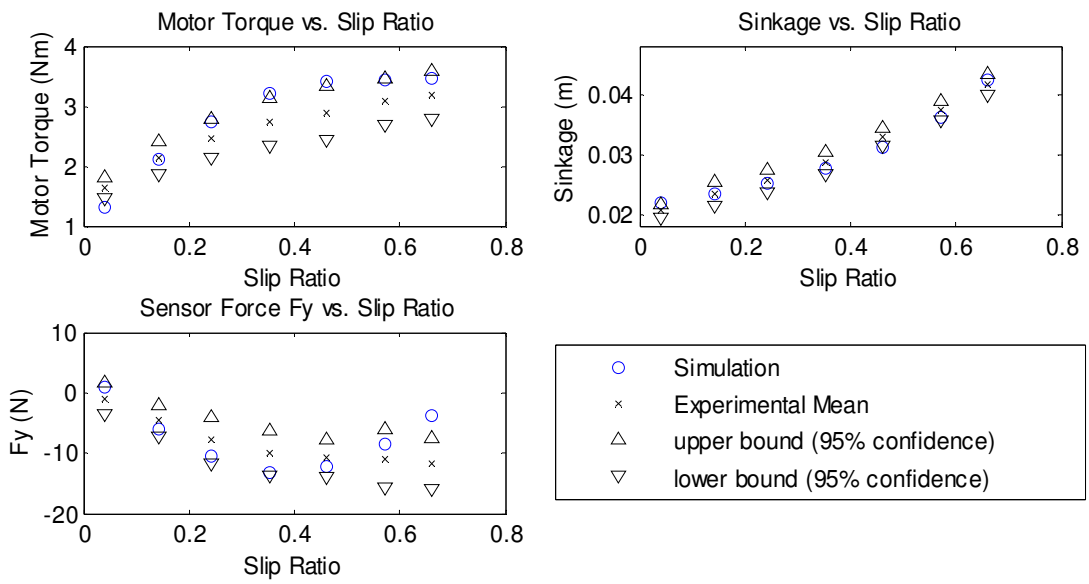

Fig. 9: Simulation and Experimental Data vs. Slip Ratio Measured Sinkage vs. Slip Ratio, 18 grousers

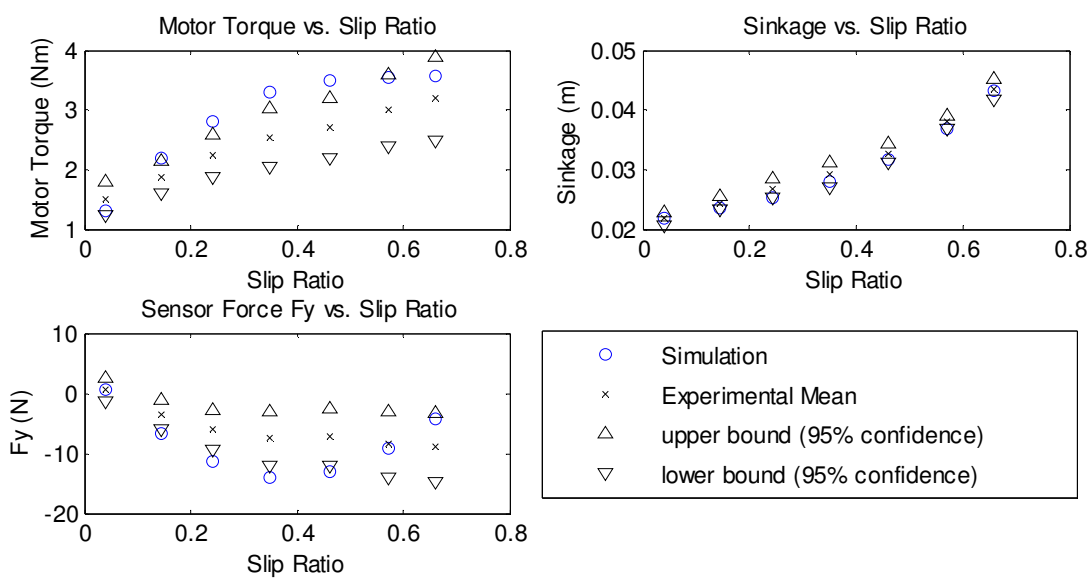

Fig. 10: Simulation and Experimental Data vs. Slip Ratio Measured Sinkage vs. Slip Ratio, 9 grousers

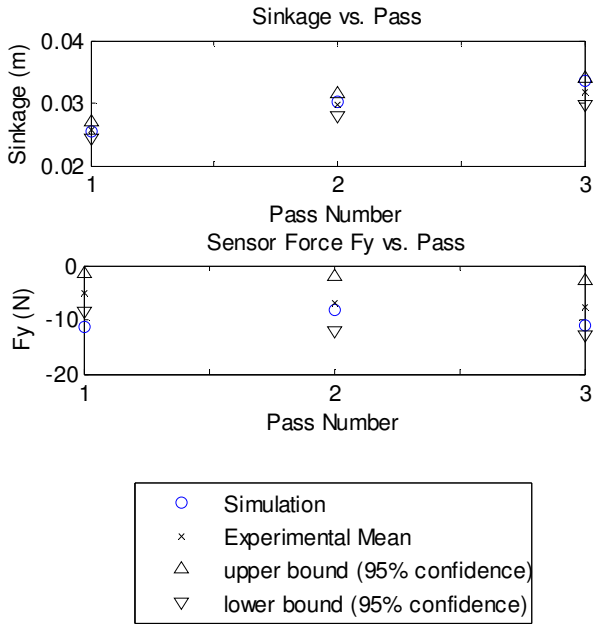

Fig. 11: Multipass Results, 9 grousers, 0.24 slip ratio

VI. Conclusions

In conclusion, experiments showed that, for the dry sandy soil used in this research, the wheel with 18 grousers had approximately 30% improvement in drawbar pull over the wheel with 9 grousers with relatively little effect on sinkage. When comparing these experimental results to simulation results, AS^{2}TM is able to capture the sinkage vs. slip ratio relationship accurately for both single and multipass cases. More research is required to further study the differences observed in the drawbar pull F_y and motor torque, as well as the ability of AS^{2}TM to model other scenarios such as side-slip.

This single-wheel research will be used to support the development of the rover dynamic simulator shown in Fig. 12 which couples a rigid multi-body dynamics engine with the AS^{2}TM wheel-soil interaction module.

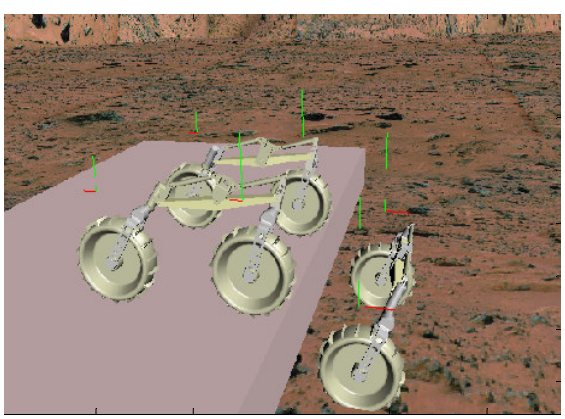

Fig. 12: Visualization of dynamic rover simulator with only suspension shown

Acknowledgements

The authors would like to thank Dr. Carsten Harnisch and Dr. Björn Lach of AESCO GbR in Hamburg for their help with and support of AS^{2}TM. The authors would like to thank Professor Steven Dubowsky, Dr. Karl Iagnemma and Dr. Christopher Brooks at the MIT Field and Space Robotics Laboratory for their help with and support of the single-wheel experiments. The authors would also like to thank Steve Fisher for designing the wheels tested in this research, as well as Robert Carr and Howard Jones for their help and support throughout the project.

References

[1] Grand, C., Ben Amar, F., Bidaud, P., Andrade, G., "A simulation system for behaviour evaluation of off-road mobile robots", *Proceedings of the 4th International Conference on Climbing and Walking Robots (CLAWAR)*, Karlsruhe, Germany, September 2001, pp. 307-314.

[2] Andrade, G., Ben Amar, F., Bidaud, P., Chatila, R., "Modeling Robot-Soil Interaction for Planetary Rover Motion Control", *Proceedings of the 1998 IEEE/RSJ International Conference on Intelligent Robots and Systems*, Victory, Canada, October 1998, pp. 576-581.

[3] Yoshiada, K., Watanabe, T., Mizuno, N., Ishigami, G., "Slip, Traction Control, and Navigation of a Lunar Rover", *Proceedings of the 7th International Symposium on Artificial Intelligence, Robotics and Automation in Space: i-SAIRAS 2003*, NARA, Japan, May 19-23, 2003.

[4] Grečenko, A., "The Slip and Drift Model of a Wheel with Tyre Compared to Some Other Attempts in this Field", *Journal of Terramechanics*, vol. 29, no. 6, 1992, pp. 559-604.

[5] The Mathworks, "SimMechanics User's Guide", 2002.

[6] AESCO, "Matlab/Simulink Module AESCO Soft Soil Tyre Model (AS^{2}TM) User's Guide", 2003.

[7] Harnisch, C., Lach, B., "Off Road Vehicles in a Dynamic Three-Dimensional Realtime Simulation", *Proceedings of the 14th International Conference of the International Society for Terrain-Vehicle Systems*, Vicksburg, MS USA, October 20-24, 2002.

[8] Schmid, I.C., "Interaction of vehicle and terrain – results from 10 years of research at IKK", Journal of Terramechanics, vol. 32, no. 1, 1995, pp. 3-26.

[9] Bekker, M.G., "Theory of Land Locomotion", The University of Michigan Press, Ann Arbor, 1956.

[10] Wong, J.Y, "Theory of Ground Vehicles", Third Edition, John Wiley & Sons, Inc., 2001.

[11] Janosi, Z. Hanamoto, B., "Analytical Determination of Drawbar Pull as a Function of Slip for Tracked Vehicles in Deformable Soils", *Proceedings of the 1st International Conference on Terrain-Vehicle Systems*, Turin, 1961.

[12] Lambe T.W., Whitman R.V. "Soil Mechanics", John Wiley & Sons, Inc., 1969.